Network Traffic Investigation Report

Project Name: SOC Project 1 – Network Traffic Investigation
Tool Used: Wireshark
Project Type: Network Traffic Analysis
Analyst: Student
Date: [10/11/2026]

1. Introduction

This project focuses on analyzing network traffic using Wireshark. A PCAP file is used to examine network packets and understand how devices communicate over a network.

The analysis includes identifying IP addresses, protocols, port numbers, and DNS queries. The main purpose is to understand network communication and learn basic network traffic investigation techniques used by a Security Operations Center (SOC) Analyst.

2. Objectives

The main objectives of this project are:

1. To analyze network traffic using Wireshark.
2. To identify DNS queries and understand their purpose.
3. To identify source and destination IP addresses.
4. To examine network protocols and port numbers.
5. To understand how network packets help in security investigations.
6. To document observations and prepare a network traffic investigation report.

3. Tools and Technologies Used

The following tools and technologies were used in this project:

- **Wireshark:** A network protocol analyzer used to capture and examine network packets.
- **PCAP File:** A file containing captured network traffic for analysis.
- **Display Filters:** Filters used to find specific types of packets.
- **DNS Protocol:** A protocol used to translate domain names into IP addresses and perform other DNS lookups.

4. PCAP File Information

Capture File Name: [SkypeIRC (1).cap]
Analysis Tool: Wireshark

Display Filter Used: dns
Selected Frame Number: 5

The PCAP file was opened in Wireshark to examine the captured network traffic. The DNS display filter was applied to identify DNS packets and study their details.

5. DNS Traffic Analysis

DNS stands for Domain Name System. It helps devices look up information associated with domain names and IP addresses.

During this investigation, the dns display filter was applied in Wireshark to display DNS packets. Frame number 5 was selected for further inspection.

The DNS packet details were examined to identify the query name, query type, source IP address, destination IP address, and port numbers.

Observed DNS Packet Details

- **Frame Number:** 5
- **Protocol:** DNS
- **Display Filter:** dns
- **DNS Query Name:** 92.in-addr.arpa
- **DNS Query Type:** PTR (as observed)
- **Source IP Address:** 192.168.1.2
- **Destination IP Address:** 192.168.1.1
- **Source Port:** 2128
- **Destination Port:** 53

6. Understanding the PTR Query

PTR stands for Pointer Record. A PTR query is commonly used for reverse DNS lookup, which looks up a domain name associated with an IP address.

The query name 92.in-addr.arpa indicates a reverse DNS lookup under the IPv4 reverse-lookup domain.

This observation demonstrates how Wireshark can be used to inspect DNS queries and understand the purpose of network traffic.

7. Wireshark Display Filter

The following display filter was used during the investigation:

dns

This filter displays DNS packets in the packet list. It helps an analyst focus on DNS traffic instead of viewing all captured packets at once.

Purpose of the filter:

- To identify DNS packets.
- To inspect DNS query names and types.
- To examine source and destination addresses.
- To understand DNS communication patterns.

8. Evidence and Screenshots

Screenshots are important because they provide visual evidence supporting the investigation findings.

The following evidence should be included:

1. Screenshot of the PCAP file opened in Wireshark.
2. Screenshot showing the dns display filter and DNS packets.
3. Screenshot showing Frame 5 and its DNS query details.

The screenshots should be saved in the project's Screenshots folder.

9. Findings

The investigation produced the following observations:

1. Wireshark was used to inspect a PCAP file.
2. The DNS display filter was applied successfully.
3. Frame number 5 was selected for detailed inspection.
4. The selected packet contained a DNS PTR query.
5. The packet details provided information about the query name, IP addresses, and port numbers.
6. The investigation demonstrated how Wireshark helps analysts examine network communication.

These observations describe the inspected packet and do not, by themselves, establish malicious activity.

10. Security Analysis

DNS traffic analysis is useful in security investigations because it helps analysts understand domain lookups and identify unusual DNS activity.

A SOC Analyst may investigate unexpected query patterns, suspicious domain names, unusual DNS volumes, or repeated failed lookups.

In this project, only the observed packet details are documented. Further investigation would be required before deciding whether any traffic is suspicious or malicious.

11. Conclusion

This project provided practical experience in using Wireshark for basic network traffic investigation.

The analysis demonstrated how to apply a DNS display filter, select an individual packet, inspect DNS query details, and document observations in a structured report.

These skills form a foundation for further learning in network security and SOC Analyst activities.

12. Future Improvements

The investigation can be extended by:

1. Analyzing additional DNS packets.
2. Examining TCP connections and port numbers.
3. Identifying frequently communicating IP addresses.
4. Comparing DNS queries to identify unusual patterns.
5. Collecting additional screenshots and documenting evidence.
6. Writing a final assessment based on the complete PCAP analysis.

13. Evidence and Screenshots

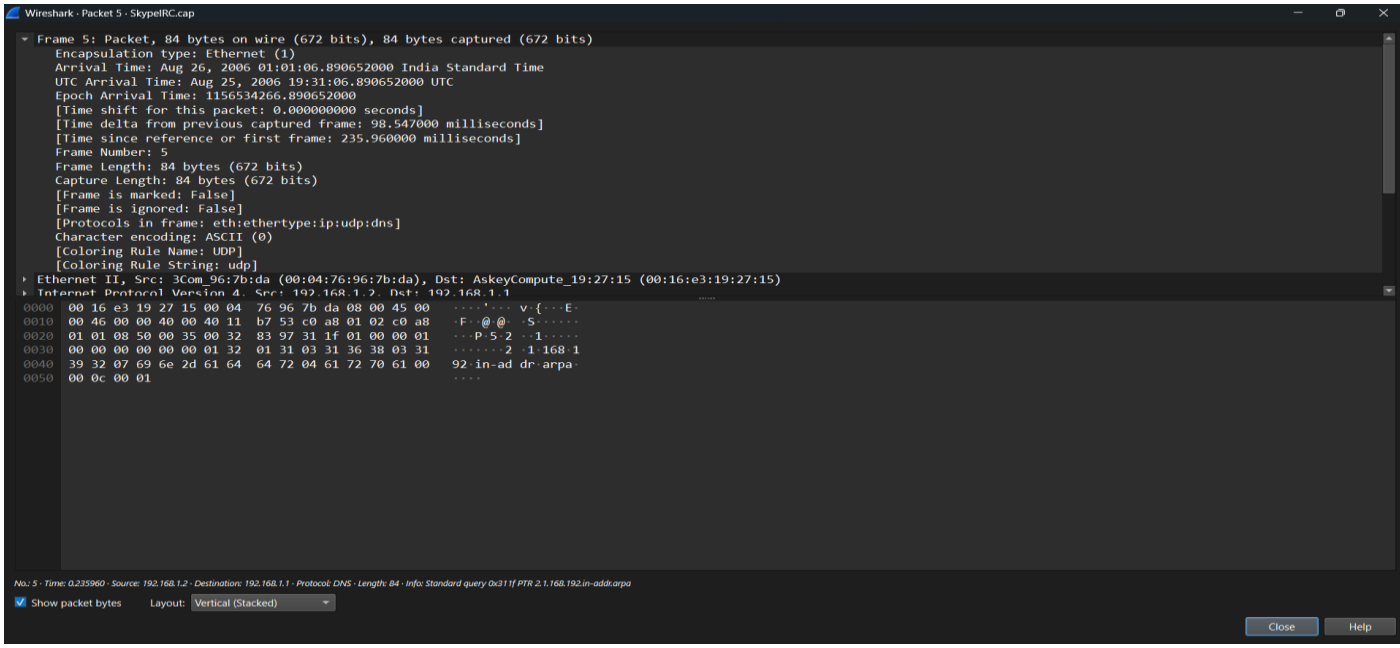
Screenshots were collected during the network traffic investigation using Wireshark. These screenshots provide visual evidence of the analysis performed.

Screenshot 1: DNS Traffic in Wireshark

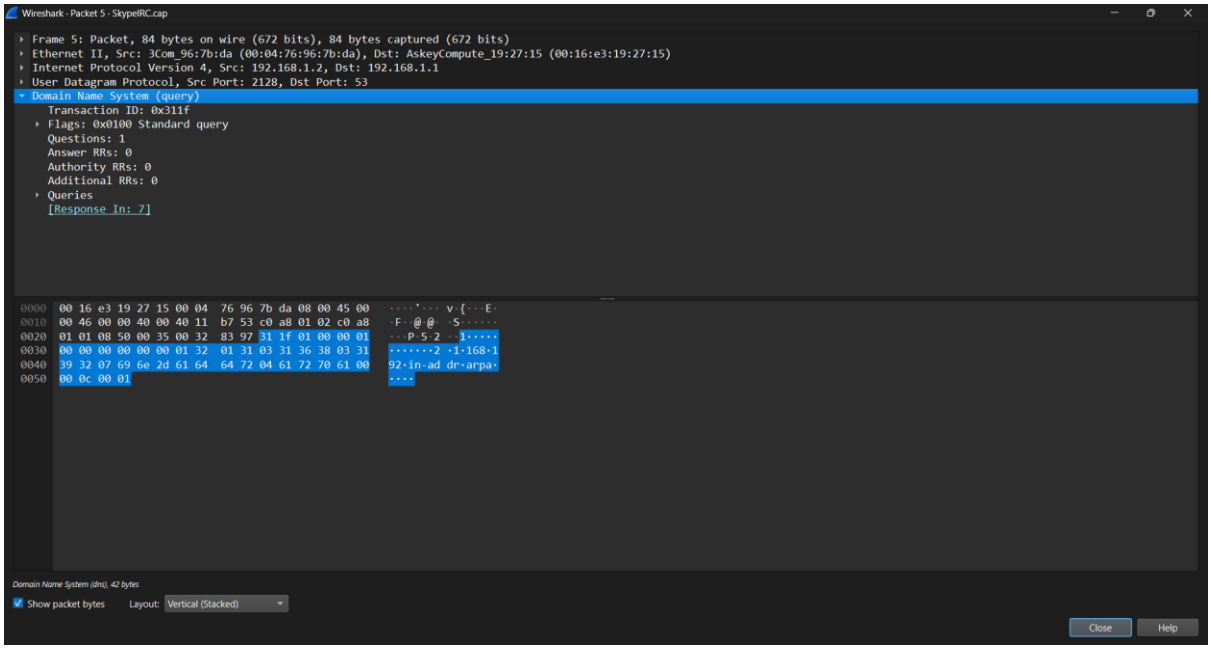
Description:

This screenshot shows DNS packets displayed in Wireshark using the display filter dns. The filter helps identify DNS traffic in the selected PCAP file.

Evidence File: DNS_Traffic.png



Screenshot 2: DNS Query Details



Description:

This screenshot shows the details of a selected DNS packet. The DNS query name and query type were examined to understand the purpose of the query.

Evidence File: DNS_Query_Details.png

14. Investigation Summary

In this project, a sample network traffic capture named SkypeIRC (1).cap was analyzed using Wireshark.

The DNS display filter was applied to identify DNS packets. Frame number 5 was selected to examine the DNS query details.

The investigation helped demonstrate how to inspect network packets, identify DNS queries, and examine IP addresses and port numbers.

15. Skills Learned

The following skills were practiced during this project:

1. Opening a PCAP file in Wireshark.
2. Applying display filters to find specific network traffic.
3. Identifying DNS packets.
4. Examining DNS query names and query types.
5. Reading source and destination IP addresses.
6. Identifying source and destination port numbers.
7. Documenting network traffic observations.
8. Preparing a basic network investigation report.

16. Limitations

This project focuses on basic DNS traffic analysis using a sample PCAP file.

Only selected packets were examined in detail. Therefore, the observations do not represent a complete analysis of every packet in the capture.

Further analysis would be required to identify broader communication patterns or determine whether any activity is suspicious.

17. Final Conclusion

This project provided practical experience in analyzing network traffic with Wireshark.

The investigation demonstrated how a SOC Analyst can use packet analysis to examine DNS queries, identify network addresses, and document findings.

The project provides a foundation for further cybersecurity learning, including TCP analysis, protocol investigation, and identifying unusual network traffic patterns.

18. TCP Traffic Analysis

Observed TCP Connection

Field	Observation
Protocol	TCP
Source IP	192.168.1.2
Destination IP	212.204.214.144
Source Port	2848
Destination Port	6667
TCP Flags	PSH, ACK

Analysis:
A TCP packet was observed from source IP 192.168.1.2 to destination IP 212.204.214.144. The source port was 2848, and the destination port was 6667. The observed TCP flags were PSH, ACK. Destination port 6667 is commonly associated with IRC traffic. However, this observation alone does not establish malicious activity.

Evidence:
A screenshot of the selected TCP packet and its expanded TCP details should be included as supporting evidence.

19. Investigation Findings

Finding 1: DNS Traffic

A DNS query for 92.in-addr.arpa was observed in Wireshark. The query type was PTR, which is used for reverse DNS lookups.

Evidence: DNS_Traffic.png and DNS_Query_Details.png

Finding 2: TCP Traffic

A TCP packet was observed from source IP 192.168.1.2 to destination IP 212.204.214.144.

- Source Port: 2848
- Destination Port: 6667
- TCP Flags: PSH, ACK

Destination port 6667 is commonly associated with IRC traffic. This observation alone does not prove malicious activity.

Evidence: TCP_Traffic_Analysis.png

Finding 3: IP Address Observations

During the IPv4 Endpoints analysis, the following entries were observed:

IP Address	Packets	Bytes
69.160.83.193	1	54
24.200.254.253	1	60
85.165.0.55	2	133

These are the entries recorded during the investigation. They should not be considered the complete list of top talkers unless all endpoints have been compared.

20. Investigation Conclusion

The PCAP file was analyzed using Wireshark to investigate network traffic. DNS and TCP packets were examined, and selected IP addresses, ports, and TCP flags were recorded.

A DNS PTR query for 92.in-addr.arpa was observed. A TCP packet was also observed from 192.168.1.2 to 212.204.214.144, using source port 2848 and destination port 6667, with PSH, ACK flags.

The analysis demonstrates how Wireshark can be used to inspect network protocols, identify IP addresses, and examine TCP packet details. The available observations alone are not sufficient to conclude that malicious activity occurred.

21. Recommended Defensive Actions

1. **Monitor network traffic:** Regularly review unusual connections and unexpected communication patterns.
2. **Investigate unfamiliar connections:** Check the purpose of unexpected IP addresses and ports before deciding whether they are suspicious.
3. **Review DNS activity:** Investigate unusual DNS queries and repeated unexpected requests.
4. **Use firewall rules carefully:** Allow only necessary network communication according to organizational security policies.
5. **Maintain evidence:** Save PCAP files, screenshots, timestamps, and investigation notes for future review.
6. **Avoid unsupported conclusions:** Confirm suspicious activity with additional evidence before labeling it malicious.